

Are there other option-pricing inconsistencies you look for?

Option models generally assume that forward prices are predictive of the future movements in the spot price. Academic research and common sense suggest that this relationship is often invalid. Forward option-pricing models can break down, particularly in interest rate markets with steep term structures and low volatility levels.

A simple example of this anomaly would be a rates trade we did in 2010. At the time, the current Brazilian interest rate was around 8 percent and the 6-month forward rate was over 12 percent. The 6-month forward option prices were distributed around the forward rate of over 12 percent. In other words, the option prices implicitly assumed the 6-month forward rate as the expected level. The implied volatility at the time was around 100 basis points normalized, which meant the market was assigning the odds of nothing happening for the next six months or so—that is, rates staying near 8 percent—as over a four-standard-deviation event.⁵

We did not have conviction about the future direction of Brazilian interest rates, much less the actual levels. But we thought the assumption that a spot rate in six months near the current spot rate was a greater than four-standard-deviation event—an assumption that was embedded in the option price—represented a mispricing. We structured a trade that had a strike price around 10 percent, which was cheap because it was well out-of-the-money based on the forward interest rate (above 12 percent), but it was actually well in-the-money based on the current interest rate (8 percent). Rates could have gone up by an amount equal to half the difference between the forward and spot rates, and we still would have made money.

Why was the market expecting such a sharp increase in Brazilian interest rates?

There were several factors. Commodity exports are a major driver in the Brazilian economy. At the time, rising commodity prices and strong growth in China (a major importer) were resulting in increased domestic inflation. There was a high conviction that the Brazilian central bank was about to initiate an extended series of very substantial rate hikes. While we certainly understood why the market saw rate hikes as likely, we